1.41 3D Matrices as Linear Transformations

Worked Solutions

1 a)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}$$

So $(2, 3, -1)$.

1 b)

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -3 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 4 \\ 15 \\ 3 \end{pmatrix}$$

So $(4, 15, 3)$.

1 c)

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix}$$

So $(1, 3, -1)$.

1 d)

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}$$

So $(1, 3, 1)$.

1 e)

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \\ 4 \end{pmatrix}$$

So $(4, 4, 4)$.

1 f)

$$\begin{pmatrix} 0 & 2 & 0 \\ -3 & -5 & 4 \\ 0 & 3 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 6 \\ -25 \\ 9 \end{pmatrix}$$

So $(6, -25, 9)$.

2 a)

This is the identity matrix. All points get mapped to themselves.

2 b)

This is the zero matrix. All points get mapped to the origin, $(0, 0, 0)$.

3)

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 4 & 1 & 3 \\ 2 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 11 & 3 & 8 \\ 6 & 1 & 4 \\ 8 & 3 & 6 \end{pmatrix}$$

$$\begin{pmatrix} a+4b+2c & a+b & a+3b+c \\ d+4e+2f & d+e & d+3e+f \\ g+4h+2i & g+h & g+3h+i \end{pmatrix} = \begin{pmatrix} 11 & 3 & 8 \\ 6 & 1 & 4 \\ 8 & 3 & 6 \end{pmatrix}$$

$$a + 4b + 2c = 11 \quad (1)$$
$$a + b = 3 \quad (2)$$
$$a + 3b + c = 8 \quad (3)$$

$\therefore a = 1,\ b = 2,\ c = 1.$

$$d + 4e + 2f = 6 \quad (4)$$
$$d + e = 1 \quad (5)$$
$$d + 3e + f = 4 \quad (6)$$

$\therefore d = 0,\ e = 1,\ f = 1.$

$$g + 4h + 2i = 8 \quad (7)$$
$$g + h = 3 \quad (8)$$
$$g + 3h + i = 6 \quad (9)$$

$\therefore g = 2,\ h = 1,\ i = 1.$ So $\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}$.

Alternative Method (using inverse matrices):

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 4 & 1 & 3 \\ 2 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 11 & 3 & 8 \\ 6 & 1 & 4 \\ 8 & 3 & 6 \end{pmatrix}$$

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = \begin{pmatrix} 11 & 3 & 8 \\ 6 & 1 & 4 \\ 8 & 3 & 6 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 4 & 1 & 3 \\ 2 & 0 & 1 \end{pmatrix}^{-1}$$

$$= \begin{pmatrix} 11 & 3 & 8 \\ 6 & 1 & 4 \\ 8 & 3 & 6 \end{pmatrix} \begin{pmatrix} 1 & -1 & 2 \\ 2 & -1 & 1 \\ -2 & 2 & -3 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}$$

4)

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 1 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 0 \\ 3 & 2 & 0 \\ 4 & 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} a+b+c & 2a+b+2c & 3a+b+c \\ d+e+f & 2d+e+2f & 3d+e+f \\ g+h+i & 2g+h+2i & 3g+h+i \end{pmatrix} = \begin{pmatrix} 2 & 1 & 0 \\ 3 & 2 & 0 \\ 4 & 1 & 1 \end{pmatrix}$$

$$a+b+c=2 \quad (1)$$
$$2a+b+2c=1 \quad (2)$$
$$3a+b+c=0 \quad (3)$$

$\therefore a=-1,\ b=3,\ c=0.$

$$d+e+f=3 \quad (4)$$
$$2d+e+2f=2 \quad (5)$$
$$3d+e+f=0 \quad (6)$$

$\therefore d=-\frac{3}{2},\ e=4,\ f=\frac{1}{2}.$

$$g+h+i=4 \quad (7)$$
$$2g+h+2i=1 \quad (8)$$
$$3g+h+i=1 \quad (9)$$

$\therefore g=-\frac{3}{2},\ h=7,\ i=-\frac{3}{2}.$ So $\begin{pmatrix} -1 & 3 & 0 \\ -\frac{3}{2} & 4 & \frac{1}{2} \\ -\frac{3}{2} & 7 & -\frac{3}{2} \end{pmatrix}$.

Alternative Method (using inverse matrices):

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = \begin{pmatrix} 2 & 1 & 0 \\ 3 & 2 & 0 \\ 4 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 1 & 2 & 1 \end{pmatrix}^{-1}$$

$$= \begin{pmatrix} 2 & 1 & 0 \\ 3 & 2 & 0 \\ 4 & 1 & 1 \end{pmatrix} \begin{pmatrix} -\frac{1}{2} & 2 & -\frac{1}{2} \\ 0 & -1 & 1 \\ \frac{1}{2} & 0 & -\frac{1}{2} \end{pmatrix} = \begin{pmatrix} -1 & 3 & 0 \\ -\frac{3}{2} & 4 & \frac{1}{2} \\ -\frac{3}{2} & 7 & -\frac{3}{2} \end{pmatrix}$$

5)

$$\begin{pmatrix} 5 & 2 & -1 \\ -1 & 0 & -1 \\ 1 & 2 & 5 \end{pmatrix} \begin{pmatrix} a & d & g \\ b & e & h \\ c & f & i \end{pmatrix} = \begin{pmatrix} 14 & -7 & -21 \\ 0 & -1 & 1 \\ -16 & 19 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 5a+2b-c & 5d+2e-f & 5g+2h-i \\ -a-c & -d-f & -g-i \\ a+2b+5c & d+2e+5f & g+2h+5i \end{pmatrix} = \begin{pmatrix} 14 & -7 & -21 \\ 0 & -1 & 1 \\ -16 & 19 & 3 \end{pmatrix}$$

$$5a+2b-c=14 \quad (1)$$
$$-a-c=0 \quad (2)$$

$$a + 2b + 5c = -16 \quad (3)$$

$\therefore a = 3,\ b = -2,\ c = -3.$

$$5d + 2e - f = -7 \quad (4)$$
$$-d - f = -1 \quad (5)$$
$$d + 2e + 5f = 19 \quad (6)$$

$\therefore d = -2,\ e = 3,\ f = 3.$

$$5g + 2h - i = -21 \quad (7)$$
$$-g - i = 1 \quad (8)$$
$$g + 2h + 5i = 3 \quad (9)$$

$\therefore g = -3,\ h = -2,\ i = 2.$ So $\begin{pmatrix} 3 & -2 & -3 \\ -2 & 3 & -2 \\ -3 & 3 & 2 \end{pmatrix}$.

Alternative Method (using inverse matrices):

$$\begin{pmatrix} a & d & g \\ b & e & h \\ c & f & i \end{pmatrix} = \begin{pmatrix} 5 & 2 & -1 \\ -1 & 0 & -1 \\ 1 & 2 & 5 \end{pmatrix}^{-1} \begin{pmatrix} 14 & -7 & -21 \\ 0 & -1 & 1 \\ -16 & 19 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{10} & -\frac{3}{5} & -\frac{1}{10} \\ \frac{1}{5} & \frac{13}{10} & \frac{3}{10} \\ -\frac{1}{10} & -\frac{2}{5} & \frac{1}{10} \end{pmatrix} \begin{pmatrix} 14 & -7 & -21 \\ 0 & -1 & 1 \\ -16 & 19 & 3 \end{pmatrix} = \begin{pmatrix} 3 & -2 & -3 \\ -2 & 3 & -2 \\ -3 & 3 & 2 \end{pmatrix}$$

6)

$$\begin{pmatrix} 2 & 1 & 2 \\ 2 & 2 & 2 \\ 1 & 2 & 1 \end{pmatrix} \begin{pmatrix} a & d & g \\ b & e & h \\ c & f & i \end{pmatrix} = \begin{pmatrix} -6 & -3 & 6 \\ -8 & -2 & 6 \\ -6 & 0 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 2a + b + 2c & 2d + e + 2f & 2g + h + 2i \\ 2a + 2b + 2c & 2d + 2e + 2f & 2g + 2h + 2i \\ a + 2b + c & d + 2e + f & g + 2h + i \end{pmatrix} = \begin{pmatrix} -6 & -3 & 6 \\ -8 & -2 & 6 \\ -6 & 0 & 3 \end{pmatrix}$$

$$2a + b + 2c = -6 \quad (1)$$
$$2a + 2b + 2c = -8 \quad (2)$$
$$a + 2b + c = -6 \quad (3)$$

$\therefore a + c = -2,\ b = -2.$

$$2d + e + 2f = -3 \quad (4)$$
$$2d + 2e + 2f = -2 \quad (5)$$
$$d + 2e + f = 0 \quad (6)$$

$\therefore d + f = -2,\ e = 1.$

$$2g + h + 2i = 6 \quad (7)$$
$$2g + 2h + 2i = 6 \quad (8)$$
$$g + 2h + i = 3 \quad (9)$$

$\therefore g + i = 3$, $h = 0$. So $b = -2$, $e = 1$, $h = 0$, $c = -2 - a$, $f = -2 - d$, $i = 3 - g$.

$$\therefore \begin{pmatrix} a & d & g \\ -2 & 1 & 0 \\ -2-a & -2-d & 3-g \end{pmatrix}$$

7 a)

Stretch, parallel to the x-axis, scale factor 5.

7 b)

Stretch, parallel to the y-axis, scale factor 4.

7 c)

Stretch, parallel to the z-axis, scale factor 6.

7 d)

Stretch, parallel to the x-axis, scale factor 7, and a stretch, parallel to the z-axis, scale factor 2.

7 e)

Enlargement, centred at the origin, scale factor 3. Alternative Method: Stretch, parallel to the x-axis, scale factor 3, and a stretch, parallel to the y-axis, scale factor 3, and a stretch, parallel to the z-axis, scale factor 3.

8)

The matrix represents a stretch, parallel to the x-axis, scale factor 2, and a stretch, parallel to the y-axis, scale factor 3. So the image will be 2×3 times the volume of the original shape. So the volume of the shape's image is $2 \times 3 \times 4 = 24$ units^3.

9)

The matrix represents an enlargement, centred at the origin, scale factor 5. So the image will be 5^3 times the volume of the original shape. So the volume of the shape's image is $5^3 \times 4 = 500$ units^3.

10)

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 10 & 0 & 0 \\ 0 & 0 & 6 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 20 & 0 & 0 \\ 0 & 0 & 30 \\ 0 & 0 & 0 \end{pmatrix}$$

The coordinates of the vertices of triangle T' are $(20, 0, 0)$, $(0, 0, 0)$ and $(0, 30, 0)$. The area of the triangle is $= \frac{1}{2} \times 20 \times 30 = 300$ units^2.

11)

$$\begin{pmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 8 & 0 & 0 \\ 0 & 0 & -4 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 24 & 0 & 0 \\ 0 & 0 & -8 \end{pmatrix}$$

The coordinates of the vertices of triangle T' are $(0, 24, 0)$, $(0, 0, 0)$ and $(0, 0, -8)$. The area of the triangle is $= \frac{1}{2} \times 24 \times 8 = 96$ units^2.

12 a)

$$\begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

12 b)

$$\begin{pmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{pmatrix}.$$

12 c)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{5} & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

12 d)

$$\begin{pmatrix} 8 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 7 \end{pmatrix}.$$

12 e)

$$\begin{pmatrix} \frac{3}{4} & 0 & 0 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & \frac{1}{8} \end{pmatrix}.$$

13)

Area of K is $\frac{7\times 4}{2} = 14$ units^2.

$$\begin{pmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{pmatrix} \begin{pmatrix} 0 & -2 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ -4 & 0 & 3 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -2a & 0 & 2a \\ 0 & 0 & 0 & 0 \\ -4a & 0 & 3a & 0 \end{pmatrix}$$

Area of K' is $\frac{7a\times 4a}{2} = 14a^2$. $14a^2 = 3 \times 14 \Rightarrow a^2 = 3 \Rightarrow a = \sqrt{3}$. So the vertices of K' have coordinates $(0, 0, -4\sqrt{3})$, $(-2\sqrt{3}, 0, 0)$, $(0, 0, 3\sqrt{3})$ and $(2\sqrt{3}, 0, 0)$.

14 a)

$$\begin{pmatrix} 4 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 9 \end{pmatrix}.$$

14 b)

$$\begin{pmatrix} 4 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 9 \end{pmatrix} \begin{pmatrix} 3m+n \\ n-m \\ m+2n \end{pmatrix} = \begin{pmatrix} 16m-n \\ -2n \\ 21m+3n \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 12m+4n \\ 8n-8m \\ 9m+18n \end{pmatrix} = \begin{pmatrix} 16m-n \\ -2n \\ 21m+3n \end{pmatrix}$$

$$12m + 4n = 16m - n \Rightarrow 4m - 5n = 0 \quad (1)$$
$$8n - 8m = -2n \Rightarrow 8m - 10n = 0 \Rightarrow 4m - 5n = 0 \quad (1)$$
$$9m + 18n = 21m + 3n \Rightarrow 12m - 15n = 0 \Rightarrow 4m - 5n = 0 \quad (1)$$
$$m + n = 9 \quad (2)$$

$\therefore m = 5,\ n = 4.$

15 a)

Reflection in the plane $y = 0$.

15 b)

Reflection in the plane $x = 0$.

15 c)

Reflection in the plane $z = 0$.

16)

The matrix represents a reflection in the plane $z = 0$. So the image will have the same volume as the original shape. So the volume of the shape's image is 4 units^3.

17)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & 3 \\ 1 & 4 & 1 \\ 1 & 1 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 3 \\ -1 & -4 & -1 \\ 1 & 1 & 4 \end{pmatrix}$$

So the vertices of T' have coordinates $(2, -1, 1)$, $(1, -4, 1)$ and $(3, -1, 4)$.

18)

$$\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 3 & 0 & -2 \\ 0 & 0 & 3 & 5 \\ 0 & 5 & 0 & 5 \end{pmatrix} = \begin{pmatrix} -3 & -3 & 0 & 2 \\ 0 & 0 & 3 & 5 \\ 0 & 5 & 0 & 5 \end{pmatrix}$$

So the vertices of X' have coordinates $(-3, 0, 0)$, $(-3, 0, 5)$, $(0, 3, 0)$ and $(2, 5, 5)$.

19 a)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

19 b)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

19 c)

$$\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

19 d)

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

19 e)

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

19 f)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

20)

$\mathbf{A}$ represents a reflection so $\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$.

$$\therefore 1 - 2k^2(2k+3) = -1 \Rightarrow 2 - 4k^3 - 6k^2 = 0 \Rightarrow 4k^3 + 6k^2 - 2 = 0$$

$$\Rightarrow 2k^3 + 3k^2 - 1 = 0 \Rightarrow (2k-1)(k+1)^2 = 0 \Rightarrow k = \frac{1}{2} \text{ or } k = -1.$$

21 a)

A rotation by 90°, anticlockwise about the x-axis.

21 b)

A rotation by 270°, anticlockwise about the y-axis.

21 c)

A rotation by 180° about the x-axis.

21 d)

A rotation by 180° about the z-axis.

21 e)

A rotation by 90°, anticlockwise about the y-axis.

21 f)

A rotation by 270°, anticlockwise about the x-axis.

21 g)

A rotation by 270°, anticlockwise about the z-axis.

21 h)

A rotation by 180° about the y-axis.

21 i)

A rotation by 90°, anticlockwise about the z-axis.

22)

The matrix represents a rotation by 90°, anticlockwise about the y-axis. So the image will have the same volume as the original shape. So the volume of the shape's image is 4 units^3.

23)

$$\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 7 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 7 \\ 1 \end{pmatrix}$$

So the coordinates of B are $(2, 7, 1)$.

24)

$$\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ -4 \\ 5 \end{pmatrix} = \begin{pmatrix} -3 \\ 4 \\ 5 \end{pmatrix}$$

So the coordinates of A are $(-3, 4, 5)$.

25)

$$\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} -1 & 2 & 4 \\ 1 & 5 & -3 \\ 3 & 1 & -2 \end{pmatrix} = \begin{pmatrix} 1 & -2 & -4 \\ 1 & 5 & -3 \\ -3 & -1 & 2 \end{pmatrix}$$

So the coordinates of the vertices of T' are $(1, 1, -3)$, $(-2, 5, -1)$ and $(-4, -3, 2)$.

26)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} p \\ q \\ r \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \Rightarrow \begin{pmatrix} p \\ r \\ -q \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

So the coordinates of P are $(a, -c, b)$.

27 a)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos 45^\circ & -\sin 45^\circ \\ 0 & \sin 45^\circ & \cos 45^\circ \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

27 b)

$$\begin{pmatrix} \cos 135^\circ & 0 & \sin 135^\circ \\ 0 & 1 & 0 \\ -\sin 135^\circ & 0 & \cos 135^\circ \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \\ -\frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

27 c)

$$\begin{pmatrix} \cos 315^\circ & -\sin 315^\circ & 0 \\ \sin 315^\circ & \cos 315^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

28 a)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos 30^\circ & -\sin 30^\circ \\ 0 & \sin 30^\circ & \cos 30^\circ \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$$

28 b)

$$\begin{pmatrix} \cos 120^\circ & 0 & \sin 120^\circ \\ 0 & 1 & 0 \\ -\sin 120^\circ & 0 & \cos 120^\circ \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \\ 0 & 1 & 0 \\ -\frac{\sqrt{3}}{2} & 0 & -\frac{1}{2} \end{pmatrix}$$

28 c)

$$\begin{pmatrix} \cos 150^\circ & -\sin 150^\circ & 0 \\ \sin 150^\circ & \cos 150^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

28 d)

A rotation by 60°, clockwise about the x-axis, is the same as a rotation by 300°, anticlockwise about the x-axis.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos 300^\circ & -\sin 300^\circ \\ 0 & \sin 300^\circ & \cos 300^\circ \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} \\ 0 & -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

28 e)

A rotation by 225°, clockwise about the y-axis, is the same as a rotation by 135°, anticlockwise about the y-axis.

$$\begin{pmatrix} \cos 135^\circ & 0 & \sin 135^\circ \\ 0 & 1 & 0 \\ -\sin 135^\circ & 0 & \cos 135^\circ \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \\ -\frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

28 f)

A rotation by 330°, clockwise about the z-axis, is the same as a rotation by 30°, anticlockwise about the z-axis.

$$\begin{pmatrix} \cos 30^\circ & -\sin 30^\circ & 0 \\ \sin 30^\circ & \cos 30^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

29)

$$\begin{pmatrix} \cos 35^\circ & 0 & \sin 35^\circ \\ 0 & 1 & 0 \\ -\sin 35^\circ & 0 & \cos 35^\circ \end{pmatrix} = \begin{pmatrix} 0.819 & 0 & 0.574 \\ 0 & 1 & 0 \\ -0.574 & 0 & 0.819 \end{pmatrix} \quad (3 \text{ SF})$$

30)

A rotation by 102°, clockwise about the x-axis, is the same as a rotation by 258°, anticlockwise about the x-axis.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos 258^\circ & -\sin 258^\circ \\ 0 & \sin 258^\circ & \cos 258^\circ \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -0.208 & 0.978 \\ 0 & -0.978 & -0.208 \end{pmatrix} \quad (3 \text{ SF})$$

31)

$$\begin{pmatrix} \cos 45^\circ & -\sin 45^\circ & 0 \\ \sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & -2 & 0 \\ 0 & 1 & 5 \\ 9 & 0 & -4 \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & -2 & 0 \\ 0 & 1 & 5 \\ 9 & 0 & -4 \end{pmatrix} = \begin{pmatrix} \frac{3\sqrt{2}}{2} & -\frac{3\sqrt{2}}{2} & -\frac{5\sqrt{2}}{2} \\ \frac{3\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & \frac{5\sqrt{2}}{2} \\ 9 & 0 & -4 \end{pmatrix}$$

So the coordinates of the vertices of T' are $\left(\frac{3\sqrt{2}}{2}, \frac{3\sqrt{2}}{2}, 9\right)$, $\left(-\frac{3\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0\right)$ and $\left(-\frac{5\sqrt{2}}{2}, \frac{5\sqrt{2}}{2}, -4\right)$.

32)

A rotation by 135°, clockwise about the x-axis, is the same as a rotation by 225°, anticlockwise about the x-axis.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos 225^\circ & -\sin 225^\circ \\ 0 & \sin 225^\circ & \cos 225^\circ \end{pmatrix} \begin{pmatrix} 0 & 2 & 2 & 0 \\ 3 & 0 & 0 & 3 \\ -1 & -1 & 2 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 0 & -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix} \begin{pmatrix} 0 & 2 & 2 & 0 \\ 3 & 0 & 0 & 3 \\ -1 & -1 & 2 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 2 & 2 & 0 \\ -2\sqrt{2} & -\frac{\sqrt{2}}{2} & \sqrt{2} & -\frac{\sqrt{2}}{2} \\ -\sqrt{2} & \frac{\sqrt{2}}{2} & -\sqrt{2} & -\frac{5\sqrt{2}}{2} \end{pmatrix}$$

So the coordinates of the vertices of R' are $(0, -2\sqrt{2}, -\sqrt{2})$, $\left(2, -\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$, $(2, \sqrt{2}, -\sqrt{2})$ and $\left(0, -\frac{\sqrt{2}}{2}, -\frac{5\sqrt{2}}{2}\right)$.

33)

As the rotation is about the z-axis, the z-coordinate of A will be the same as the z-coordinate of A'.

$$\begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 16 \\ -64 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ k \\ 4 \end{pmatrix} \Rightarrow \begin{pmatrix} 16\cos\theta + 64\sin\theta \\ 16\sin\theta - 64\cos\theta \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ k \\ 4 \end{pmatrix}$$

$$16\cos\theta + 64\sin\theta = 0 \Rightarrow 16 + \frac{64\sin\theta}{\cos\theta} = 0 \Rightarrow 16 + 64\tan\theta = 0 \Rightarrow \tan\theta = -\frac{1}{4}.$$

As θ is obtuse, we must have $\sin\theta = \frac{1}{\sqrt{17}}$ and $\cos\theta = -\frac{4}{\sqrt{17}}$.

$$\text{So } k = 16\left(\frac{1}{\sqrt{17}}\right) - 64\left(-\frac{4}{\sqrt{17}}\right) = 16\sqrt{17}$$

$\therefore A'(0, 16\sqrt{17}, 4)$.

34)

A rotation by 45°, clockwise about the x-axis, is the same as a rotation by 315°, anticlockwise about the x-axis.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos 315^\circ & -\sin 315^\circ \\ 0 & \sin 315^\circ & \cos 315^\circ \end{pmatrix} \begin{pmatrix} 1 & 0 & 2 \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ k\sqrt{2} & -\frac{3\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 0 \\ 2 & -2 & -1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & 0 & 2 \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 0 & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \begin{pmatrix} 1 & 0 & 2 \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ k\sqrt{2} & -\frac{3\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 0 \\ 2 & -2 & -1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & 0 & 2 \\ k & -1 & 0 \\ k & -2 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 0 \\ 2 & -2 & -1 \end{pmatrix} \Rightarrow k = 2.$$

35 a)

$\cos\theta = -\frac{1}{2} \Rightarrow \theta = \{120^\circ, 240^\circ\}$. $\sin\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = \{60^\circ, 120^\circ\}$. So it is a rotation by 120°, anticlockwise about the x-axis.

35 b)

$\cos\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = \{30^\circ, 330^\circ\}$. $\sin\theta = \frac{1}{2} \Rightarrow \theta = \{30^\circ, 150^\circ\}$. So it is a rotation by 30°, anticlockwise about the z-axis.

35 c)

$\cos\theta = \frac{\sqrt{2}}{2} \Rightarrow \theta = \{45^\circ, 315^\circ\}$. $\sin\theta = -\frac{\sqrt{2}}{2} \Rightarrow \theta = \{225^\circ, 315^\circ\}$. So it is a rotation by 315°, anticlockwise about the y-axis.

35 d)

$\cos\theta = \frac{\sqrt{6}+\sqrt{2}}{4} \Rightarrow \theta = \{15°, 345°\}$. $\sin\theta = \frac{\sqrt{2}-\sqrt{6}}{4} \Rightarrow \theta = \{195°, 345°\}$. So it is a rotation by 345°, anticlockwise about the y-axis.

35 e)

$\cos\theta = \frac{\sqrt{6}+\sqrt{2}}{4} \Rightarrow \theta = \{15°, 345°\}$. $\sin\theta = \frac{\sqrt{6}-\sqrt{2}}{4} \Rightarrow \theta = \{15°, 165°\}$. So it is a rotation by 15°, anticlockwise about the z-axis.

36)

$\cos\theta = \frac{1}{3} \Rightarrow \theta = \{70.5°, 289°\}$ (3 SF). $\sin\theta = \frac{2\sqrt{2}}{3} \Rightarrow \theta = \{70.5°, 109°\}$ (3 SF). So $\theta = 70.5°$ (3 SF).

37)

$\cos\theta = \frac{\sqrt{7}}{4} \Rightarrow \theta = \{48.6°, 311°\}$ (3 SF). $\sin\theta = \frac{3}{4} \Rightarrow \theta = \{48.6°, 131°\}$ (3 SF). So $\theta = 48.6°$ (3 SF).

38)

$\cos\theta = \frac{\sqrt{6}-\sqrt{2}}{4} \Rightarrow \theta = \{75°, 285°\}$. $\sin\theta = \frac{\sqrt{6}+\sqrt{2}}{4} \Rightarrow \theta = \{75°, 105°\}$. So $\theta = 75°$.

$$75n = 360k \Rightarrow 3 \times 5^2 \times n = 2^3 \times 3^2 \times 5 \times k \Rightarrow n = 2^3 \times 3 \times 5^{-1} \times k.$$

Smallest integer value for n when $k = 5$, so $n = 24$.

39)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} -3 \\ 4 \\ 7 \end{pmatrix} = \begin{pmatrix} -3 \\ -7 \\ -4 \end{pmatrix} \Rightarrow \begin{pmatrix} -3 \\ 4\cos\theta - 7\sin\theta \\ 4\sin\theta + 7\cos\theta \end{pmatrix} = \begin{pmatrix} -3 \\ -7 \\ -4 \end{pmatrix}$$

$$4\cos\theta - 7\sin\theta = -7 \qquad (1)$$
$$4\sin\theta + 7\cos\theta = -4 \qquad (2)$$

$\therefore \cos\theta = -\frac{56}{65}$ and $\sin\theta = \frac{33}{65}$. $\cos\theta = -\frac{56}{65} \Rightarrow \theta = \{149°, 211°\}$ (3 SF). $\sin\theta = \frac{33}{65} \Rightarrow \theta = \{30.5°, 149°\}$ (3 SF). So $\theta = 149°$ (3 SF).

40)

$$\begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix} \begin{pmatrix} 1 \\ 8 \\ -6 \end{pmatrix} = \begin{pmatrix} -6 \\ 8 \\ 1 \end{pmatrix} \Rightarrow \begin{pmatrix} \cos\theta - 6\sin\theta \\ 8 \\ -\sin\theta - 6\cos\theta \end{pmatrix} = \begin{pmatrix} -6 \\ 8 \\ 1 \end{pmatrix}$$

$$\cos\theta - 6\sin\theta = -6 \qquad (1)$$
$$-\sin\theta - 6\cos\theta = 1 \qquad (2)$$

$\therefore \cos\theta = -\frac{12}{37}$ and $\sin\theta = \frac{35}{37}$. $\cos\theta = -\frac{12}{37} \Rightarrow \theta = \{109°, 251°\}$ (3 SF). $\sin\theta = \frac{35}{37} \Rightarrow \theta = \{71.1°, 109°\}$ (3 SF). So $\theta = 109°$ (3 SF).

41)

$$\begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 4 \\ 8 \\ 2 \end{pmatrix} = \begin{pmatrix} -8 \\ -4 \\ 2 \end{pmatrix} \Rightarrow \begin{pmatrix} 4\cos\theta - 8\sin\theta \\ 4\sin\theta + 8\cos\theta \\ 2 \end{pmatrix} = \begin{pmatrix} -8 \\ -4 \\ 2 \end{pmatrix}$$

$$4\cos\theta - 8\sin\theta = -8 \qquad (1)$$
$$4\sin\theta + 8\cos\theta = -4 \qquad (2)$$

$\therefore \cos\theta = -\frac{4}{5}$ and $\sin\theta = \frac{3}{5}$. $\cos\theta = -\frac{4}{5} \Rightarrow \theta = \{143°, 217°\}$ (3 SF). $\sin\theta = \frac{3}{5} \Rightarrow \theta = \{36.9°, 143°\}$ (3 SF). So $\theta = 143°$ (3 SF).

42)

$$\begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix} \begin{pmatrix} 5 \\ -5 \\ -6 \end{pmatrix} = \begin{pmatrix} 6 \\ -5 \\ -5 \end{pmatrix} \Rightarrow \begin{pmatrix} 5\cos\theta - 6\sin\theta \\ -5 \\ -5\sin\theta - 6\cos\theta \end{pmatrix} = \begin{pmatrix} 6 \\ -5 \\ -5 \end{pmatrix}$$

$$5\cos\theta - 6\sin\theta = 6 \qquad (1)$$
$$-5\sin\theta - 6\cos\theta = -5 \qquad (2)$$

$\therefore \cos\theta = \frac{60}{61}$ and $\sin\theta = -\frac{11}{61}$. $\cos\theta = \frac{60}{61} \Rightarrow \theta = \{10.4°, 350°\}$ (3 SF). $\sin\theta = -\frac{11}{61} \Rightarrow \theta = \{190°, 350°\}$ (3 SF). So $\theta = 350°$ (3 SF) about the y-axis.

43 a)

$$\mathbf{BA} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} -1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & -1 \end{pmatrix} = \begin{pmatrix} -2 & 0 & 2 \\ -1 & 2 & 1 \\ -1 & 2 & 1 \end{pmatrix}$$

43 b)

$$\mathbf{AB} = \begin{pmatrix} -1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{pmatrix} = \begin{pmatrix} -2 & 1 & -1 \\ 0 & 2 & -2 \\ 2 & -1 & 1 \end{pmatrix}$$

44)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

So this is a rotation by 180° about the y-axis.

45 a)

Rotation by 90°, anticlockwise about the x-axis.

45 b)

$\mathbf{A}^2$ is a rotation by 180°, anticlockwise about the x-axis. $\mathbf{A}^3$ is a rotation by 270°, anticlockwise about the x-axis. $\mathbf{A}^4$ is a rotation by 360°, anticlockwise about the x-axis. $\mathbf{A}^5$ is a rotation by 90°, anticlockwise about the x-axis. $\mathbf{A}^6$ is a rotation by 180°, anticlockwise about the x-axis. $\mathbf{A}^7$ is a rotation by 270°, anticlockwise about the x-axis. $\mathbf{A}^8$ is a rotation by 360°, anticlockwise about the x-axis. ... $\frac{2025}{4} = 506$ r 1. So $\mathbf{A}^{2025}$ is a rotation by 90°, anticlockwise about the x-axis.

46 a)

Rotation by 270°, anticlockwise about the z-axis.

46 b)

$\mathbf{A}^2$ is a rotation by 180°, anticlockwise about the z-axis. $\mathbf{A}^3$ is a rotation by 90°, anticlockwise about the z-axis. $\mathbf{A}^4$ is a rotation by 360°, anticlockwise about the z-axis. $\mathbf{A}^5$ is a rotation by 270°, anticlockwise about the z-axis. $\mathbf{A}^6$ is a rotation by 180°, anticlockwise about the z-axis. $\mathbf{A}^7$ is a rotation by 90°, anticlockwise about the z-axis. $\mathbf{A}^8$ is a rotation by 360°, anticlockwise about the z-axis. ... $\frac{443}{4} = 110$ r 3. So $\mathbf{A}^{443}$ is a rotation by 90°, anticlockwise about the z-axis.

47 a)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{pmatrix} \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 15 \end{pmatrix}$$

47 b)

$$\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

47 c)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -5 \\ 0 & 1 & 0 \end{pmatrix}$$

47 d)

$$\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

47 e)

$$\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

48)

$$\begin{pmatrix} -k & m & 0 \\ \frac{4k}{m} & k & 0 \\ 0 & 0 & -k\sqrt{3} \end{pmatrix}^2 = \begin{pmatrix} 12 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 12 \end{pmatrix} \Rightarrow \begin{pmatrix} k^2+4k & 0 & 0 \\ 0 & 4k+k^2 & 0 \\ 0 & 0 & 3k^2 \end{pmatrix} = \begin{pmatrix} 12 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 12 \end{pmatrix}$$

$k^2+4k=12 \Rightarrow k=-6$ or $k=2$. $3k^2=12 \Rightarrow k=\pm 2$. $\therefore k=2$.

49)

$$\begin{pmatrix} 6 & 3 & 4 \\ -2 & 2 & 4 \\ 5 & 4 & 1 \end{pmatrix} \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 6a+3d+4g & 6b+3e+4h & 6c+3f+4i \\ -2a+2d+4g & -2b+2e+4h & -2c+2f+4i \\ 5a+4d+g & 5b+4e+h & 5c+4f+i \end{pmatrix} = \begin{pmatrix} -5 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

$$6a+3d+4g=-5 \quad (1)$$
$$-2a+2d+4g=0 \quad (2)$$
$$5a+4d+g=0 \quad (3)$$

$\therefore a=-\frac{7}{9}$, $d=\frac{11}{9}$, $g=-1$.

$$6b+3e+4h=0 \quad (4)$$
$$-2b+2e+4h=-5 \quad (5)$$
$$5b+4e+h=0 \quad (6)$$

$\therefore b=\frac{13}{18}$, $e=-\frac{7}{9}$, $h=-\frac{1}{2}$.

$$6c+3f+4i=0 \quad (7)$$
$$-2c+2f+4i=0 \quad (8)$$
$$5c+4f+i=5 \quad (9)$$

$\therefore c=-\frac{2}{9}$, $f=\frac{16}{9}$, $i=-1$.

$$a=-\frac{7}{9},\ b=\frac{13}{18},\ c=-\frac{2}{9},\ d=\frac{11}{9},\ e=-\frac{7}{9},\ f=\frac{16}{9},\ g=-1,\ h=-\frac{1}{2},\ i=-1$$

So $\begin{pmatrix} -\frac{7}{9} & \frac{13}{18} & -\frac{2}{9} \\ \frac{11}{9} & -\frac{7}{9} & \frac{16}{9} \\ -1 & -\frac{1}{2} & -1 \end{pmatrix}$.

Alternative Method (using inverse matrices):

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = \begin{pmatrix} 6 & 3 & 4 \\ -2 & 2 & 4 \\ 5 & 4 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{7}{45} & -\frac{13}{90} & -\frac{2}{45} \\ -\frac{11}{45} & \frac{7}{45} & \frac{16}{45} \\ \frac{1}{5} & \frac{1}{10} & -\frac{1}{5} \end{pmatrix} \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -\frac{7}{9} & \frac{13}{18} & -\frac{2}{9} \\ \frac{11}{9} & -\frac{7}{9} & \frac{16}{9} \\ -1 & -\frac{1}{2} & -1 \end{pmatrix}$$

50 a)

$$\begin{pmatrix} 4 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 4 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Stretch, parallel to the y-axis, scale factor 3, followed by an enlargement, centred at the origin, scale factor 4.

50 b)

$$\begin{pmatrix} 6 & 0 & 0 \\ 0 & -6 & 0 \\ 0 & 0 & 6 \end{pmatrix} = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Reflection in the plane $y = 0$, followed by an enlargement, centred at the origin, scale factor 6.

50 c)

$$\begin{pmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

A rotation by 180° about the z-axis, followed by an enlargement, centred at the origin, scale factor 2.

50 d)

$$\begin{pmatrix} 0 & 8 & 0 \\ -8 & 0 & 0 \\ 0 & 0 & 8 \end{pmatrix} = \begin{pmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

A rotation by 270° anticlockwise about the z-axis, followed by an enlargement, centred at the origin, scale factor 8.